

13/05/2025

Task-2-1

Implement Conditional, control and looping statements.

Aim: To implement Conditional, control and looping statements using python.

2.1: You are developing a single grade management system for a school. The system needs to determine the grade of a student based on their score.

If the score is 90 or above, the grade is 'A'.

If the score is between 80 and 89 the grade is 'B'.

If the score is between 70 and 69, the grade is 'D'.

If the score is between 70 and 79, the grade is 'C'.

If the score is below 60, the grade is 'F'.

Algorithm:

1. Start
2. Get the input mark from the user
3. With the use of an if-elif-else statement do
 - If the marks ≥ 90 print grade 'A'.
 - If the mark is between 80 and 89 print grade 'B'.
 - If the mark is between 70 and 79 print grade 'C'.
 - If the mark is b/w 60 and 69 print grade 'D'.
 - If the mark is below 60, and print grade 'F'.
4. Stop.

Program:

```
score = int(input("Enter the score:"))
```

```
if score  $\geq$  90;
```

```
    printf("The grade is A")
```

```
elif (score  $\leq$  89 and score  $\geq$  80);
```

```
    printf("The grade is B")
```

```
elif (score  $\leq$  79 and score  $\geq$  70);
```

```
    printf("The grade is C")
```

```
elif (score  $\leq$  69 and score  $\geq$  60);
```

```
    printf("The grade is D")
```

```
else:
```

```
    printf("The grade is F")
```

output

Restart: c:\use

Enter the score: 60

The grade is D.

EX. NO.	TECH. - ONE
PERFORMANCE (5)	
RESULT AND ANALYSIS (5)	
VIVA VOCE (3)	
RECORD (4)	
TOTAL (17)	
DATE	

2.2 The electronics maintenance team at a data Center needs a tool to assess the health status of ups backup batteries based on their current charge percentage. You are asked to develop a python program that accepts the battery charge percentage as input and categorizes the battery health using the following conditions.

~~→ If the percent:~~

Task: write a python program that uses ladderized if-elif-else statements.

Algorithm:

1. Accept battery percentage from the user.
2. Use ladderized if-elif-else to determine the health category:
 - If percentage $\geq 90 \rightarrow$ "Excellent Battery Health".
 - If $70 \leq$ percentage $< 90 \rightarrow$ "Good Battery health".
 - If $40 \leq$ percentage $< 70 \rightarrow$ "Average Battery health".
 - If percentage $< 40 \rightarrow$ "poor battery health".

program:

```
# Battery health checker
percentage = int(input("Enter battery percentage:"))
if percentage >= 90:
    print("Excellent Battery Health")
elif percentage >= 70:
    print("Good Battery Health")
elif percentage >= 40:
    print("Average Battery Health")
else:
    print("poor battery Health")
```



output:

Enter battery percentage: 85

Good battery health.



2.3 You're coding a system at an amusement park that checks that height of each visitor.

- If the height is 120cm or more, print "Allowed".
- Otherwise, print, "Not allowed".

Repeat this for 5 visitors.

Algorithm start the program.

1. set the total number of visitors to 5;
2. Loop from visitor 1 to visitor 5:
 - Accept the height of the visitor as input (in cm)
 - If height is greater than or equal to 120, print "Allowed".
 - Else, print "Not allowed".
3. End the loop after 5 visitors have been checked
4. stop the program.

Program:

```
for i in range(1,6):  
    height = int(input(f"Enter height of visitor {i} in cm:"))  
    if height >= 120:  
        print("Allowed to ride.")  
    else:  
        print("Not allowed to ride.")
```

VEL TECH	
EX NO.	
PERFORMANCE (5)	2
RESULT AND ANALYSIS (5)	3
VIVA VOCE (5)	3
RECORD (5)	3
TOTAL (20)	
DATE WITH DATE	

Result: Thus, the python program was successfully implemented using conditional statements (if-else), control flow and looping statements)

The electronics manufacturer team at a local electronics store is looking to assess the health status of user backup batteries based on their current charge percentage. You are asked to develop a Python program that accepts the battery charge percentage as input and categorizes the battery into one of the following conditions.

Write a Python program that uses laboratory data to categorize the battery health status.

Read battery percentage from the user. The program should categorize the battery health status based on the percentage.

Output:

Allowed

Not Allowed

Allowed

Not Allowed

Allowed

if battery health checkers percentage = int(input("Enter battery percentage: "))

if percentage >= 90:

print("Excellent Battery Health")

elif percentage >= 70:

print("Good Battery Health")

elif percentage >= 50:

print("Average Battery Health")

else:

print("Poor Battery Health")